15|

$U + (V - W) =$

$$\begin{bmatrix} 5.5 \\ 33 \\ -11 \end{bmatrix}$$

because matrix addition is commutative and associative

$C + Ow =$

undefined Cannot add matrices of sizes 3×1 and 3×2

$OE + u - V =$

undefined Cannot add matrices of sizes 3×1 and 3×2

15|

$$u=\begin{bmatrix}1.5\\0\\-3.0\end{bmatrix} \quad v=\begin{bmatrix}-1\\3\\2\end{bmatrix} \quad w=\begin{bmatrix}-5\\-30\\10\end{bmatrix} \quad E=\begin{bmatrix}0&2\\3&4\\3&-1\end{bmatrix} \quad C=\begin{bmatrix}5&2\\-2&4\\1&0\end{bmatrix}$$

$$(u+v)-w =$$

$$\left(\begin{bmatrix}1.5\\0\\-3.0\end{bmatrix}+\begin{bmatrix}-1\\3\\2\end{bmatrix}\right)-\begin{bmatrix}-5\\-30\\10\end{bmatrix} =$$

$$\left(\begin{bmatrix}0.5\\3\\-1\end{bmatrix}\right)-\begin{bmatrix}-5\\-30\\10\end{bmatrix} =$$

$$\begin{bmatrix}0.5\\3\\-1\end{bmatrix}+\begin{bmatrix}5\\30\\-10\end{bmatrix} =$$

$$\begin{bmatrix}5.5\\33\\-11\end{bmatrix}$$

11)

$$A=\begin{bmatrix}4 & -2 & 3\\ -2 & 1 & 6\\ 1 & 2 & 2\end{bmatrix} \quad B=\begin{bmatrix}1 & -3 & 0\\ -3 & 1 & 0\\ 0 & 0 & -2\end{bmatrix}$$

$AB =$

$$\begin{bmatrix}4(1)+-2(-3)+3(0) & 4(-3)+-2(1)+3(0) & 4(0)+-2(0)+3(-2)\\ -2(1)+1(-3)+6(0) & -2(-3)+1(1)+6(0) & -2(0)+1(0)+6(-2)\\ 1(1)+2(-3)+2(0) & 1(-3)+2(1)+2(0) & 1(0)+2(0)+2(-2)\end{bmatrix}$$

$$\boxed{\begin{bmatrix}10 & -14 & -6\\ -5 & 7 & -12\\ -5 & -1 & -4\end{bmatrix}}$$

$AB^T =$ because $B^T = B$

$AB =$

$$\boxed{\begin{bmatrix}10 & -14 & -6\\ -5 & 7 & -12\\ -5 & -1 & -4\end{bmatrix}}$$

11

$BA =$

$$\begin{bmatrix} 1(4)+-3(-2)+0(1) & 1(-2)+-3(1)+0(2) & 1(3)+-3(6)+0(2) \\ -3(4)+1(-2)+0(1) & -3(-2)+1(1)+0(2) & -3(3)+1(6)+0(2) \\ 0(4)+0(-2)+-2(1) & 0(-2)+0(1)+-2(2) & 0(3)+0(6)+-2(2) \end{bmatrix} =$$

$$\begin{bmatrix} 10 & -5 & -15 \\ -14 & 7 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

$B^TA = BA$ because $B^T = B$

$BA =$

$$\begin{bmatrix} 10 & -5 & -15 \\ -14 & 7 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

3]

$$\begin{aligned} x+y-z &= 9 \\ 8y+6z &= -6 \\ -2x+4y-6z &= 40 \end{aligned} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ -2 & 4 & -6 & 40 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ 0 & 6 & -8 & 58 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 4 & 3 & -3 \\ 0 & 3 & -4 & 29 \end{array}\right] \Rightarrow$$

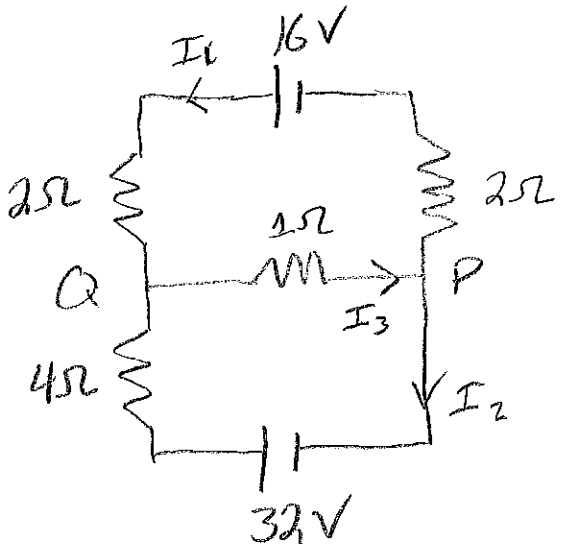
$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 1 & 7 & -32 \\ 0 & 3 & -4 & 29 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 41 \\ 0 & 1 & 7 & -32 \\ 0 & 0 & -25 & 125 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 41 \\ 0 & 1 & 7 & -32 \\ 0 & 0 & 1 & -5 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -5 \end{array}\right] \Rightarrow \boxed{x=1,\ y=3,\ z=-5}$$

17



Node P $-i_1 - i_2 + i_3 = 0$

Node Q $i_1 + i_2 - i_3 = 0$

$4i_1 + i_3 = 16$

$4i_2 + i_3 = 32$

$$\left[\begin{array}{ccc|c} -1 & -1 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ 4 & 0 & 1 & 16 \\ 0 & 4 & 1 & 32 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -4 & 5 & 16 \\ 0 & 4 & 1 & 32 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -4 & 5 & 16 \\ 0 & 0 & 6 & 48 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -4 & 5 & 16 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 8 \\ 0 & -4 & 0 & -24 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 8 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \boxed{i_1 = 2A,\ i_2 = 6A,\ i_3 = 8A}$$